

Disharee Bhowmick

Data Scientist | Machine Learning · Time-Series Forecasting · Computer Vision · MLOps

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Summary

Data scientist with an M.S. in Computer Science and 5+ years building and deploying machine learning in Python, from predictive models and time-series forecasting to computer vision and LLMs. At Cognizant, took models from prototype to production on Azure and AWS and cut recurring IT incidents by **25%** with operations teams. Published a time-series method that shrinks model inputs up to **30x** while retaining **85–90%** of accuracy (*Pattern Recognition Letters*, 2026). Eager to apply these methods to manufacturing, demand forecasting, and process optimization.

Professional Experience

Graduate Research Assistant, Machine Learning & Computer Vision

Summer 2024 – present

Auburn University, Dept. of Computer Science & Software Engineering

Auburn, AL

- Cut time-series input size by up to **30x** while retaining **85–90%** of full-resolution accuracy across **150+** classification and forecasting datasets, by designing STaTS, an unsupervised change-detection method in Python/PyTorch (first author, *Pattern Recognition Letters*, 2026).
- Enabled video models to recognize and localize actions outside their training categories by adapting Vision-Language Models to temporal segmentation.
- Estimated livestock respiration from video without contact sensors by co-developing a computer vision pipeline for smart-agriculture monitoring (IEEE MWSCAS, 2026).

Cognizant Technology Solutions

Sep 2019 – Jun 2023

Programmer Analyst Trainee → Programmer Analyst → Associate

Pune, India

Completed Cognizant's graduate training in data analysis, statistics, and Python (Sep 2019 – Feb 2020).

Generative AI (Mar 2022 – Jun 2023)

- Deployed LLM solutions to production in client workflows, integrated with downstream scoring logic, by serving open-source LLMs on **AWS SageMaker** and GPT models on **Azure OpenAI** with **Docker**, **Git**, and **CI/CD**.
- Improved answer accuracy and reduced hallucinations on enterprise knowledge bases, validated through RAG evaluation, by building RAG pipelines (chunking, embeddings, hybrid search) and fine-tuning LLMs on domain data.
- Met enterprise compliance standards by setting up **model monitoring**, logging, lineage tracking, and PII redaction.

Machine Learning, Internal Project (Mar 2021 – Feb 2022)

- Led development of a predictive analytics model on **100K+** records, selecting the final approach by benchmarking supervised and ensemble models built with scikit-learn and pandas.

Data Analytics (Mar 2020 – Feb 2021)

- Reduced recurring IT incidents by **25%** for enterprise clients by pinpointing root causes in incident data with **SQL** and **Python**, working with operations teams on fixes, and tracking before/after impact in Power BI and Tableau.

Skills

- **ML & Statistics:** Predictive modeling, time-series forecasting, anomaly detection, supervised & unsupervised learning, XGBoost, CatBoost, ensembles, feature engineering, model evaluation, hypothesis testing, physics-informed & hybrid modeling
- **Computer Vision & GenAI:** CNNs, Transformers, Vision-Language Models, video analysis, LLM fine-tuning, RAG
- **MLOps & Cloud:** Databricks, Azure ML, MLflow, model deployment & monitoring, Docker, Kubernetes, CI/CD, Git (version control), code review, Agile, ML governance; Azure OpenAI, AWS SageMaker
- **Languages & Data:** Python, SQL, scikit-learn, PyTorch, TensorFlow, pandas, polars, NumPy, Spark/PySpark, data engineering, data warehousing, IoT/sensor data, Power BI, Tableau

Education

M.S., Computer Science, Auburn University | GPA 3.83

Aug 2023 – May 2025

B.Tech., Computer Science, University of Engineering and Management, Kolkata | GPA 3.6

Aug 2015 – Jul 2019

Publications

- **Bhowmick, D.**, Ramanathan, R., Aakur, S. N. STaTS: Structure-aware temporal sequence summarization via statistical window merging. *Pattern Recognition Letters*, 202, 135–140, 2026.
- Palem, R. R., Garcia-Ascolani, M. E., **Bhowmick, D.**, et al. Vision-Based Livestock Respiratory Monitoring for Smart Agriculture Sensing Systems. *IEEE MWSCAS*, 2026.
- Co-authored preprints: CVT-Bench, multimodal LLM evaluation (arXiv:2603.21114, 2026); video-based livestock respiration estimation.

Awards & Teaching

- 100+ Women Strong Outstanding Graduate Award (2026; \$2,000), Samuel Ginn College of Engineering; sole CSSE nominee, ACM SIGHPC Fellowship; Rising Star and Star of the Month awards, Cognizant.
- Graduate Teaching Assistant, Auburn: Lead TA, Algorithms; TA, Formal Languages; Lab Instructor, Intro to Engineering.